



Kubernetes Advanced Interview Notes

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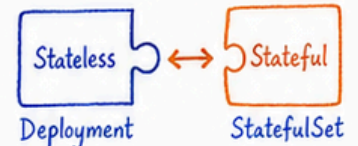
Advanced Questions & Short Answers

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1. Core Workloads

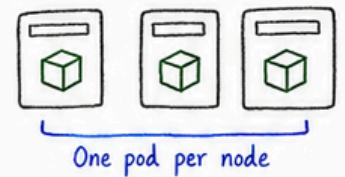
① Deployment vs StatefulSet —

- Deployment for stateless apps; StatefulSet for stateful apps with stable pod identity, ordered rollout, and persistent storage.



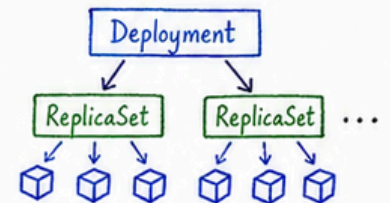
② DaemonSet —

- runs one pod on all or selected nodes; used for logging, monitoring, networking.



③ ReplicaSet vs Deployment —

- ReplicaSet maintains replica count; Deployment manages ReplicaSets and supports rolling updates and rollback.



④ What happens when a node fails? —

- control plane marks node unavailable; workloads are recreated on healthy nodes depending on controller and timing.



⑤ etcd —

- distributed key-value store holding cluster state and config.



⑥ Why etcd is critical —

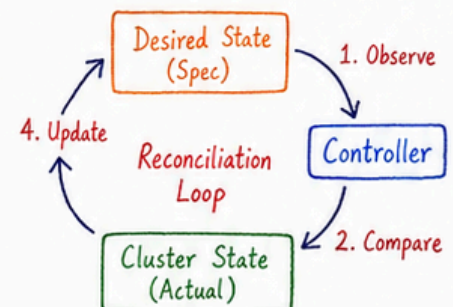
- without it, cluster desired state cannot be reliably recovered.



No etcd,
No Cluster
Recovery!

⑦ Reconciliation loop —

- controllers compare desired vs current state and act to make them match.



⑧ kube-scheduler —

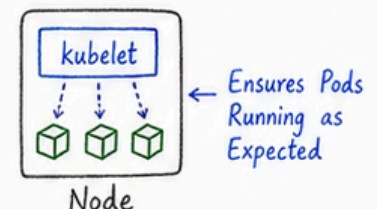
- chooses the best node for unscheduled pods.

⑨ kube-controller-manager —

- runs controllers such as node, replica, endpoints, job.

⑩ kubelet —

- node agent that ensures pod containers run as expected.





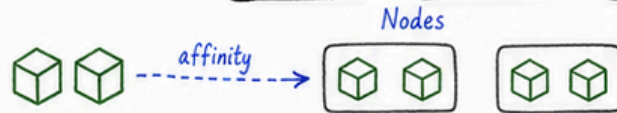
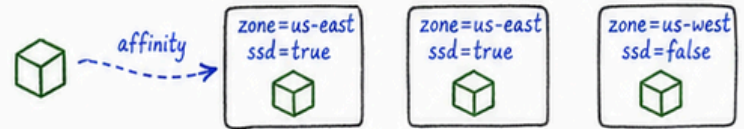
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2. Scheduling

- (11) **Node Affinity** —
 - schedules pods to nodes based on labels.
- (12) **Pod Affinity** —
 - places pods near other matching pods.
- (13) **Pod Anti-Affinity** —
 - keeps matching pods apart for high availability.
- (14) **Taints & Tolerations** —
 - taints repel pods unless matching tolerations exist.
- (15) **NodeSelector vs Node Affinity** —
 - nodeSelector is simple exact match; affinity is more flexible with rules such as In, NotIn, Exists, preferred.
- (16) **TopologySpreadConstraints** —
 - spreads pods across nodes/zones/regions for better availability.



Taint on Node
 key = dedicated
 value = gpu
 effect = NoSchedule



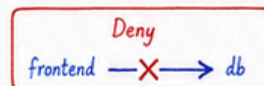
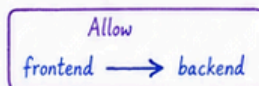
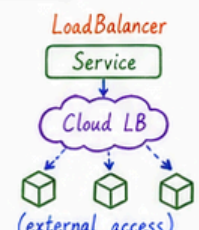
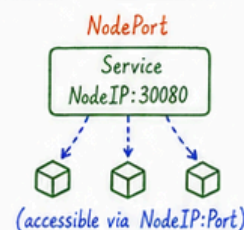
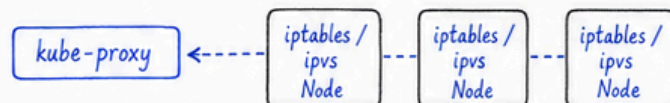
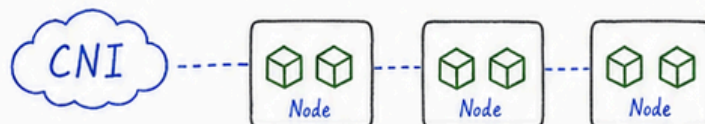
nodeSelector
 disktype: ssd
 env: prod
 (exact match)

Node Affinity
 matchExpressions:
 - key: disktype
 operator: In
 values: [ssd, nvme]
 (flexible rules)



3. Networking

- (17) **CNI** —
 - Container Network Interface; provides pod networking. Examples: Calico, Cilium, Flannel.
- (18) **Pod-to-pod communication** —
 - every pod gets an IP; pods can communicate without NAT between pods.
- (19) **kube-proxy** —
 - implements Service networking on nodes.
- (20) **ClusterIP vs NodePort vs LoadBalancer** —
 - ClusterIP internal; NodePort exposes a port on nodes; LoadBalancer requests external load balancer.
- (21) **Headless Service** —
 - clusterIP: None; used for direct pod discovery, common with StatefulSets.
- (22) **NetworkPolicy** —
 - controls allowed traffic between pods and sometimes external endpoints.
- (23) **Is NetworkPolicy enforced automatically?** —
 - No, the CNI plugin must support policy enforcement.



→ Allowed
 -X→ Denied





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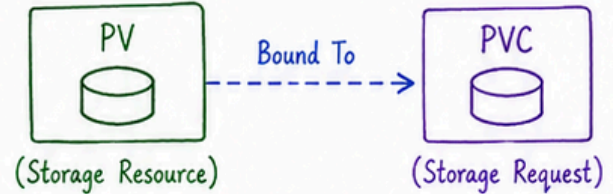
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4. Storage

(24) PV vs PVC —

- PV is storage resource; PVC is a workload's storage request.



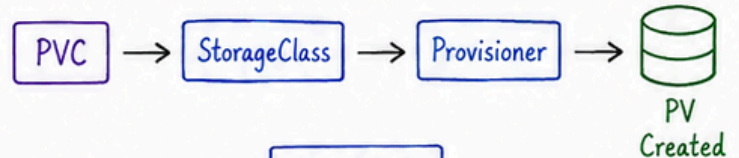
(25) StorageClass —

- defines how dynamic storage should be provisioned.



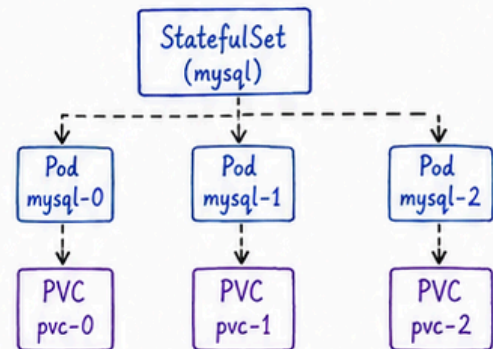
(26) Dynamic volume provisioning —

- Kubernetes automatically creates storage when a PVC uses a suitable StorageClass.



(27) StatefulSet volumeClaimTemplates —

- creates a separate PVC for each pod.
e.g., mysql-0 → pvc-0, mysql-1 → pvc-1, mysql-2 → pvc-2.



5. Health Checks

(28) Liveness vs Readiness —

- liveness decides restart; readiness decides whether pod gets traffic.



(29) Startup Probe —

- for slow-starting apps; prevents liveness/readiness from interfering during startup.



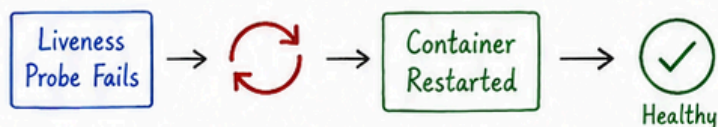
(30) What if readiness probe fails? —

- container keeps running, but pod becomes NotReady and is removed from Service endpoints.



(31) What if liveness probe fails? —

- kubelet restarts the container.





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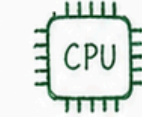
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6. Resources & Scaling

32) Request vs Limit —

- request is used for scheduling;
- limit is the maximum allowed resource usage where enforceable.



request = 250m



limit = 500m

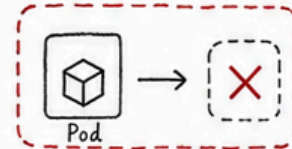
request → for scheduling
 limit → max allowed (enforceable)

33) Exceed memory limit —

- may be terminated as OOMKilled.



OOMKilled



34) Exceed CPU limit —

- usually throttled, not killed.

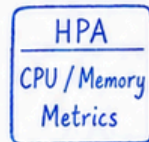


CPU

Throttled (not killed)

35) HPA —

- Horizontal Pod Autoscaler changes replica count based on metrics.



Increase replica count based on metrics

36) VPA —

- Vertical Pod Autoscaler recommends or adjusts CPU/memory requests.



cpu: 250m
mem: 256Mi

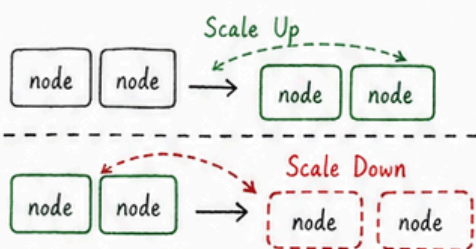


cpu: 500m
mem: 512Mi

Adjusts requests up or down

37) Cluster Autoscaler —

- adds or removes worker nodes based on scheduling demand and safe scale-down.



☆ Requests = guaranteed baseline, Limits = upper cap ☆

HPA vs VPA vs Cluster Autoscaler	What it scales	What it changes	Decision based on
HPA	Pods (replicas)	Number of Pods	CPU / Memory / Custom Metrics
VPA	Pod resources	CPU / Memory requests & limits	Usage patterns
Cluster Autoscaler	Nodes (workers)	Number of Nodes	Scheduling demand & safe scale-down



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7. Security

- (38) RBAC — Role-Based Access Control
- defines who can do what.



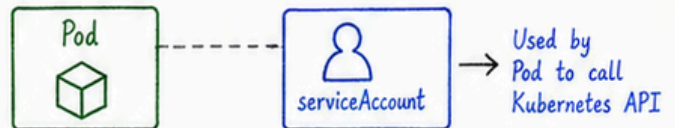
- (39) Role vs ClusterRole —
- Role is namespace-scoped;
 - ClusterRole is cluster-wide or reusable across namespaces.



- (40) RoleBinding vs ClusterRoleBinding —
- RoleBinding grants permissions in a namespace;
 - ClusterRoleBinding grants cluster-wide permissions.



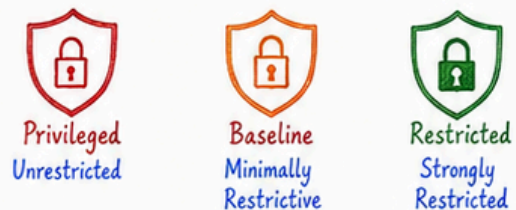
- (41) ServiceAccount —
- identity for workloads inside Kubernetes.



- (42) SecurityContext —
- security settings like
 - runAsNonRoot: true
 - readOnlyRootFilesystem: true
 - allowPrivilegeEscalation: false



- (43) Pod Security Admission —
- built-in enforcement of Pod Security Standards: Privileged, Baseline, Restricted.

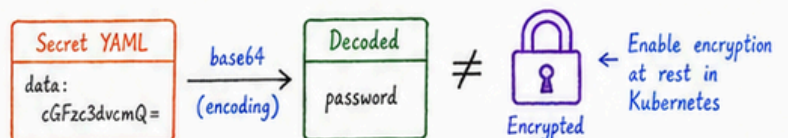


8. Configuration

- (44) ConfigMap vs Secret —
- ConfigMap for non-sensitive config;
 - Secret for sensitive data.



- (45) Are Secrets encrypted by default? —
- No; base64 is encoding, not encryption; encryption at rest should be configured.



- (46) How apps consume ConfigMaps/Secrets —
- via environment variables, command arguments, or mounted volumes.





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9. Deployment Strategies

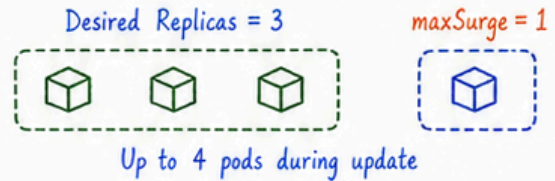
47 Rolling Update —

- gradually replaces old pods with new ones while keeping availability.



48 maxSurge —

- extra pods allowed above desired replicas during update.



49 maxUnavailable —

- how many pods may be unavailable during update.



50 Rollback a Deployment —

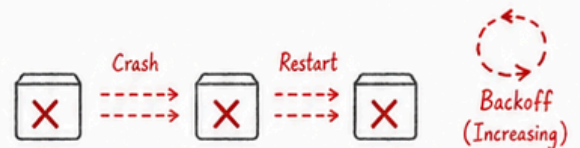
- kubectl rollout undo deployment/myapp
- history: kubectl rollout history deployment/myapp



10. Troubleshooting

51 CrashLoopBackOff —

- container keeps crashing and restarting with backoff.



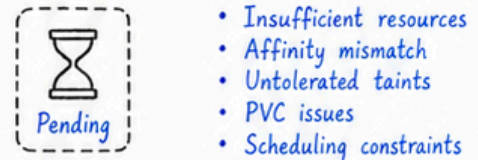
52 ImagePullBackOff —

- image pull failed; causes include bad image name/tag, auth issue, registry or network issue.



53 Why a pod stays Pending —

- insufficient resources, affinity mismatch, untolerated taints, PVC issues, scheduling constraints.



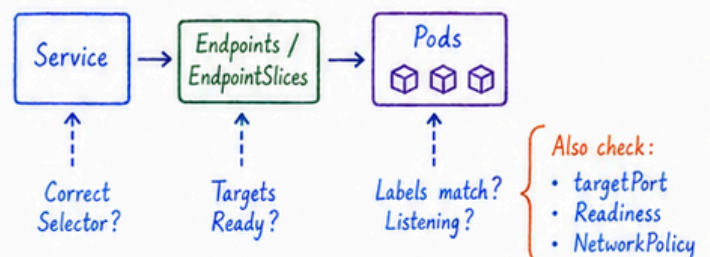
54 OOMKilled —

- container was killed due to memory pressure/limit exceedance.



55 How to troubleshoot a Service —

- check svc, endpoints/endpointslices, pod labels, targetPort, readiness, NetworkPolicy.



Useful Commands

- kubectl logs <pod>
- kubectl logs <pod> --previous
- kubectl describe pod <pod>
- kubectl get svc
- kubectl get endpoints
- kubectl get endpointslices



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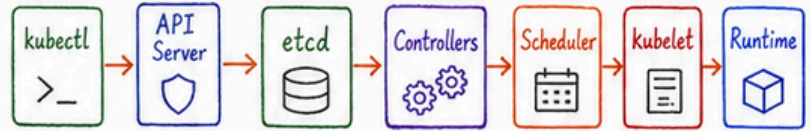
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11. Architecture

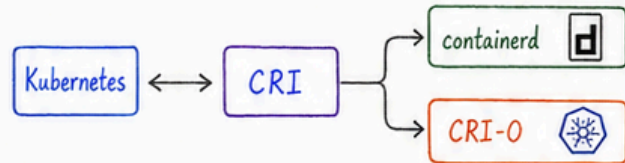
56) What happens after kubectl apply? —

- API Server validates request, stores state in etcd, controllers reconcile, scheduler assigns pods, kubelet starts containers via runtime.



57) CRI —

- Container Runtime Interface; lets Kubernetes talk to runtimes like containerd and CRI-O.



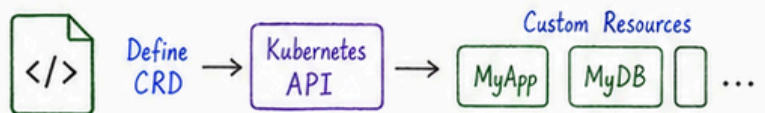
58) CSI —

- Container Storage Interface; integrates storage systems with Kubernetes.



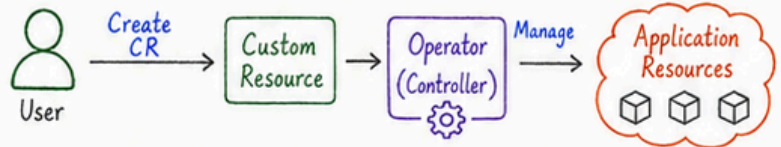
59) CRD —

- CustomResourceDefinition extends the Kubernetes API with custom resources.



60) Operator —

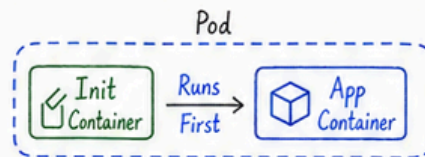
- uses custom resources and controllers to automate app-specific operations.



12. Advanced Concepts Part 1

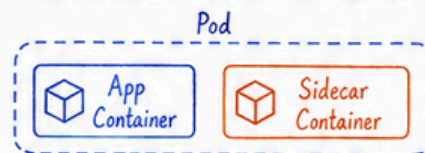
61) Init Container —

- runs and completes before app containers start.



62) Sidecar Container —

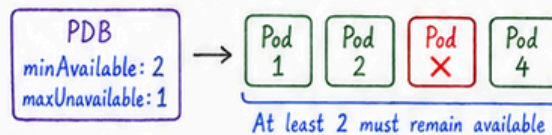
- helper container for logging, proxying, config sync, etc.



- Logging
- Proxying
- Config Sync
- ...

63) PodDisruptionBudget —

- limits unavailable replicas during voluntary disruptions.



At least 2 must remain available

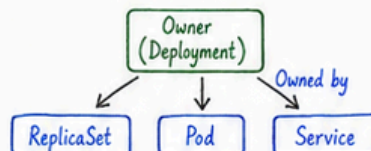
64) Finalizer —

- delays deletion until cleanup is complete.



65) OwnerReference —

- defines ownership between resources and supports garbage collection.



If owner is deleted, dependents are garbage collected.



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12. Advanced Concepts Part 2

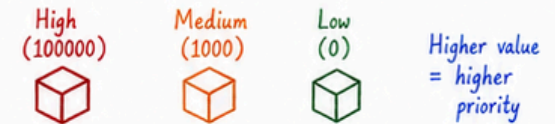
66) Garbage Collection —

- cleans up dependent resources based on ownership and deletion policy.



67) PriorityClass —

- assigns pod priority.



68) Pod Preemption —

- scheduler may evict lower-priority pods for a higher-priority pod.



69) Admission Controllers —

- validate or mutate API requests before persistence.



70) Mutating vs Validating Admission —

- mutating can modify objects; validating allows or rejects.



13. Scenario Questions

71) Pod is Running but app is unreachable —

- check readiness, Service selector, Service port/targetPort, EndpointSlice, Ingress/Gateway/LoadBalancer, NetworkPolicy, app logs.



72) Pod works by IP but not by Service name —

- check Service config; selector, EndpointSlices, CoreDNS, DNS config, network plugin.



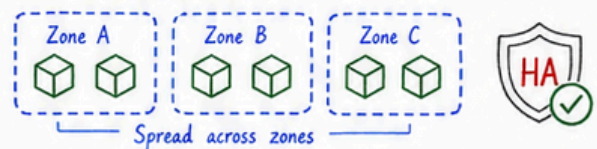
73) Pods not scheduling on a new node —

- check node readiness, taints, resources, affinity, node selectors, events.



74) How to design high availability —

- use multiple replicas, multiple nodes, zone spread, pod anti-affinity or topology spread, readiness probes, PodDisruptionBudget, autoscaling.



75) Investigate high CPU usage —

- check kubectl top pods, kubectl top nodes, describe pod, logs, CPU requests/limits, app behavior, HPA, monitoring metrics.



★ Top 10 Must-Know Advanced Questions ★

- How does Kubernetes schedule a pod?
- What happens when you run kubectl apply?
- Difference between liveness, readiness, startup probes?
- What are requests vs limits and how do they work?
- Difference between taints/tolerations and affinity?
- How does Service networking work in Kubernetes?
- How would you troubleshoot a Pending pod?
- Difference between Deployment, StatefulSet, DaemonSet?
- How does Kubernetes handle node failure?
- How would you design a high availability application?

